

Estimation of Muscle Cross-Sectional Area (ACSA) from Ultrasound Images using Feature-Based Machine Learning

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Abstract—Segmentation of ultrasound images is crucial for analysing muscle structure in sports and medical applications. Manual segmentation is tedious and may vary from person to person. In this project, we propose a simple machine learning approach in order to segment muscle regions from ultrasound images. The method includes feature extraction and selection, preprocessing, and model training. A set of important features are chosen based on their usefulness and consistency. These features help in capturing key patterns in the image. The results show that the model is able to segment muscle regions with reasonable stability and accuracy. The approach is efficient, simple and useful for basic medical image analysis tasks.

Index Terms—ultrasound segmentation, muscle imaging, random forest, feature engineering, CLAHE, ACSA estimation, classical machine learning, biomedical image analysis

I. INTRODUCTION

Rehabilitation and critical care In sports medicine, ultrasound can be used often to assess the look of muscles, and to measure the amount of muscle mass an athlete has. One of the measurements is known as Atlas Cross-Sectional Area (ACSA). Evokes the greater part of the muscle that is essentially a cut through of a transverse scan that offers clinicians an attainable sense of muscle volume and general functionality.. It is a cumbersome exercise to egoseat it by hand, and prone to inter-assessor differences as well as not being scalable. This is the reason why automated segmentation is worth undertaking.

It is an extension of a pixel-by-pixel classification pipeline of ultrasound muscle classification. We started with an earlier code (midsem code), followed by a series of particular experiments to optimise the quality of the model generalisation to new images, a spatial-Feature ablation experiment to figure out whether the model was learning actual visual features, and proceeded to refine the results with further feature, sampling, augmentation, probability smoothing and threshold selection experiments. The last (this version) version has a mean Dice of 0.751 on the test set.

II. LITERATURE REVIEW

A. Ultrasound Segmentation

Noble and Boukerroui [1] give a thorough overview of ultrasound image segmentation, covering everything from simple thresholding, statistical shape methods, and region growing to deformable models. They point out to the fact that speckle noise and missing boundaries make it harder for general

purpose segmentation techniques to work reliably, which is why domain-specific feature design helps.

B. Classical ML for Medical Imaging

Criminisi *et al.* [2] demonstrated that in medical images random forests are efficient in pixel-level labelling, in part due to the capable multi-scale intensity and context in providing the classifier with sufficient signal to differentiate anatomical structures. Haralick *et al.* [3] texture properties of grey level co-occurrence matrices (GLCM) are since then a ubiquitous technique used in the analysis of ultrasound due to its sensitivity to the speckle properties and tissue echo-structure representative of tissue types.

C. Deep Learning for Segmentation

The encoder-decoder architecture that became the default in medical image segmentation is U-Net proposed by Ronneberger *et al.* [4]. Depending on the amount of data, deep models usually perform better than classical approaches, but, as Litjens *et al.* [5] observe, the size of the dataset and the quality of annotations and limitations in computation capabilities frequently render deep learning not an option in practice in a clinical scenario.

D. Muscle Segmentation Specifically

Classical and learned methods The muscle segmentation technique of ultrasound had been approached both classically and through the method of learning. Cunningham *et al.* applied active contours along with intensity-based initialization to tibialis anterior segmentation to achieve about 0.80 with Dice scores on a small dataset [6]. Recently published work using deep learning has reported larger values of Dice, although on much less expansion sizes, than can be realistically furnished by most clinical research.

The pipeline, which is outlined in this paper, lies in a viable middle space: it is trained on a dataset of less than 200 images, does not incur the computational cost of deep learning, and its competitive performance is obtained through a combination of smart engineering, instead of complexity.

III. DATA AND METHODS

A. Dataset Description

The data consist of B-mode ultrasound images of the rectus abdominis muscle, obtained at the University Hospital Basel (DSBG).

A binary segmentation mask annotated by hand by an expert, and ground-truth ACSA values (in cm²) in a structured spreadsheet (Results.xlsx) are also provided.

The failed-images.txt list contained a subset of images which indicated samples that contained artefacts during acquisition or were missing their corresponding masks.

The remaining pairs of image masks were divided at the image level into training (70%), validation (15%), and test (15%) sets. A fixed random seed (42) was used to ensure reproducibility.

B. Preprocessing

Every picture is subjected to identical preprocessing procedures regardless of the split:

- Scan region cropping: Columns that have very low average intensity (less than a threshold of 10) are eliminated at the left and right sides of the scan to eliminate the ultrasound interface and only keep the scan region.
- Greyscale conversion: The conversion of images to 8-bit greyscale is done when needed.
- Median filtering: A 5×5 median filter removes speckle noise and maintains edge structure.
- CLAHE: Contrast Limited Adaptive Histogram Equalisation (clip 2.0, grid 8×8) is used to localise contrast and make the muscle boundaries more visible.
- Normalisation: This is the scaling of the pixel values to the range $[0, 1]$.

C. Feature Extraction

Each pixel is represented by a 20-dimensional feature vector which captures the local appearance, texture, and spatial context:

- Multi-scale mean and standard deviation: Computed over 7×7 , 21×21 , 51×51 , and 71×71 windows. These store local brightness and texture-scale variation.
- Local entropy and variance: Local entropy and local variance are calculated in a radius of 15 pixels and give the values of local texture complexity.
- Intensity ratio: The division of the pixel value by its local mean, which contains relative brightness.
- Edge features: Laplacian and gradient magnitude ($\sigma = 3$) indicate edges and transitions between tissue types.
- GLCM characteristics: Contrast, homogeneity, energy, and correlation calculated on 32×32 patches at various orientations, and averaging of the results gets rotation-invariant features.
- Spatial features: Distance to image centre and distance to image boundary, as soft anatomical priors.

Four low-importance features of the previous versions, fine-scale gradients (Gx, Gy), gradient magnitude at $\sigma = 1$, and standard deviation at 7×7 , were eliminated using feature importance analysis.

This reduced the feature vector to 20 dimensions and slightly improved generalisation.

D. Model / Approach

A Random Forest classifier with 200 decision trees is trained on the extracted features.

Gini impurity is used to grow trees with a maximum depth of 20 and a minimum of 4 samples per leaf node.

Rather than making a direct prediction of class labels, probability maps are created.

To obtain a probability map, `predict_proba` is used.

This has the advantage of allowing the decision threshold to be adjusted rather than being fixed at 0.5.

A Gaussian filter ($\sigma = 2$) is used to smooth the probability map, minimise local noise, and enhance spatial coherence.

This is thereafter thresholded to give a binary mask.

The last few postprocessing operations, such as morphological closing, hole filling, and connected component filtering, yielded the final segmentation.

E. Experimental Setup

1) *Pixel Sampling Strategy*: The training is done with stratified pixel sampling to handle the computation limitations as well as the imbalance between classes.

Each training image is sampled with 1500 pixels of muscular tissue, uniformly distributed across the mask, and 2000 negative pixels across a band at its edges (5–50 pixels surrounding the mask), and 500 more background pixels across the rest of the region.

This sampling approach, centred on boundaries, ensures the classifier is trained to differentiate between muscle and surrounding tissues, especially at the edges of the muscles where errors are most likely to occur.

2) *Data Augmentation*: Training images are augmented with intensity-based augmentation to enhance the ability to resist any changes in acquisition conditions.

The images are randomly distorted by brightness and contrast scaling (range $[0.85, 1.15]$) and additive Gaussian noise (standard deviation 2–5).

This works well to enhance the diversity of the training data without changing the spatial structure.

Validation and test images are not augmented.

3) *Threshold Tuning*: The probability map is turned into a binary mask with a threshold that is chosen on the validation set.

The search is coarsely swept up and down candidate thresholds, then narrowly around the best value.

The maximum threshold that maximises the mean Dice score on the validation set is chosen.

The best threshold in this model was 0.540.

4) *ACSA Calibration*: The physical area can be estimated by converting the number of pixels to cm² with a training set calibration factor.

This factor is calculated as the ground-truth ACSA divided by the associated pixel area, with the median used to minimise sensitivity to outliers.

This is only a rough calibration, since the imaging depth and scaling can vary across scans.

5) *Evaluation Metrics*: Dice coefficient and Intersection-over-Union (IoU), and pixel-level accuracy are used to measure segmentation performance. Findings are presented in the validation and test sets as summary statistics (mean and median).

IV. RESULTS AND DISCUSSION

A. Experiment-wise Results

1) *Spatial Feature Ablation (Diagnostic Experiment)*:

We experimented with this version, as we purposefully removed the two spatial features (`dist_centre` and `dist_boundary`) of the feature set of the midsem code before finalising this version. In their absence, the mean Dice among the validation set decreased to approximately 0.62-0.65 compared to approximately 0.77. This collapse demonstrates that the model has a dependence on spatial position, albeit to a smaller degree, but the fact that it can still achieve a score above 0.62 when only appearance features are used confirms that it is truly learning about what muscle tissue looks like. Using this we retained the spatial features as a beneficial positional prior.

TABLE I
SPATIAL FEATURE ABLATION RESULTS (VALIDATION SET)

Configuration	Dice (mean)	IoU (mean)
Midsem code (with spatial features)	≈ 0.77	≈ 0.66
Ablation (no spatial features)	≈ 0.63	≈ 0.47

2) *Spatial Feature Ablation (Diagnostic Experiment)*:

We experimented with this version, as we purposefully removed the two spatial features (`dist_centre` and `dist_boundary`) of the feature set of the midsem code before finalising this version. In their absence, the mean Dice among the validation set decreased to approximately 0.62-0.65 compared to approximately 0.77. This collapse demonstrates that the model has a dependence on spatial position, albeit to a smaller degree, but the fact that it can still achieve a score above 0.62 when only appearance features are used confirms that it is truly learning about what muscle tissue looks like. Using this we retained the spatial features as a beneficial positional prior.

The decrease in performance is factual, but not high in nature, which underlines the fact that the appearance-dependent features (and multi-scale statistics, GLCM, gradients, etc.) do have the actual discriminative information. Preservation of the spatial features is in line with the anatomical fact that the muscle easily is close to the centre of a properly cropped ultrasound field.

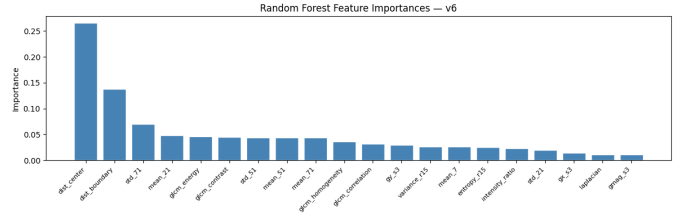


Fig. 1. Feature Importance

3) *Final Pipeline (This Version) Results*: Table II shows the overall segmentation metrics for this version across validation and test sets.

TABLE II
THIS VERSION SEGMENTATION PERFORMANCE (THRESHOLD = 0.540)

Split	Dice (mean)	IoU (mean)	Accuracy (mean)
Validation	0.777	0.661	0.965
Test	0.751	0.641	0.962

The difference between validation and test performance is rather small, implying the pipeline generalises well with the unseen images. These high accuracy scores (greater than 0.96) are partly due to the imbalance of the classes in pixel-level segmentation a classifier which identifies all the pixels as background would also have a high accuracy score but would receive a Dice score of zero. This is why the numbers that are important in this case are Dice and IoU.

B. Analysis and Insights

1) *Qualitative Performance*: Just looking at the predicted masks directly (see Section V), it is also evident that there are two distinct performance levels. When the muscle is imaged in the transverse plane with a high-quality fascial delimitation - such as in the case of the Dice = 0.938, IoU = 0.883 picture called `rectus_img_23.tif` - the predicted mask is smooth and clean with close ground truth adherence. The predicted mask is more likely to be fragmented when the muscle is observed at an oblique angle, when the muscle is framed by tendons or when the muscle is sitting in an unnatural position - such as in the `rectus_img_223.tif` and `rectus_img_346.tif` images (both Dice = 0.667, IoU = ≈ 0.50). This implies that the existing range of features may be delicate to the angle of view and fat layer size.

2) *Feature Importance*: According to the Random Forest feature importances, it is clear that the highest importance (0.263) goes to the feature, which is the most important in terms of its ranking, i.e., `dist_centre` and the feature with the second highest importance (0.135) is the most crucial, i.e., `dist_boundary`. Of the appearance measures standard deviation at large scale, measured by large-scale standard deviation (log mean) (computed as:), is next in order, then medium scale standard deviation through mean intensity (`mean_21`, 0.050). Subtle gradient features and the Laplacian rank appear at the bottom justifying the trimming of them in the feature set in this run. The general impression is that the most important

two factors in the classifier model are: a) typically the position of the muscle close to the centre of the image, and b) the typical texture of the muscle at a coarse scale.

3) *ACSA Estimation*: The predicted muscle area in cm² is the predicted area that is closely associated with the clinical ground truth of the validation set and the test set respectively with correlation coefficient of $r = 0.576$ and $r = 0.680$ respectively. What is apparent is that there is an overestimation of smaller muscles (ACSA below circa 10 cm²) and this is also probably explained by the impact of sampling strategy (boundary-focused) where the predicted mask is biased towards a slightly large value at the edge. When using larger muscles (ACSA bigger than 20 cm²) the estimates are smaller and the pipeline performs well.

C. Comparison with Previous Approaches

Results of this version are put into perspective in Tables three and four, comparing them to the midsem code.

The version is somewhat more pruned on Dice and IoU than the midsem code, courtesy of a feature pruning (cutting off the low-importance fine-scale gradient features), sampling more heavily on the boundary, and probability smoothing. The difference to deep learning is actual, yet that type of methods are generally based on 5-10 times more training data and cannot be directly equated in the same conditions. When there are less than 200 images, the Random Forest pipeline is competitive and can be practically deployed.

TABLE III
COMPARISON WITH PRIOR APPROACHES (VALIDATION)

Method	Dice	IoU	Validation Dataset
Midsem code (RF, 24 features)	0.760	0.648	Same
This version (20 features)	0.777	0.661	Same

TABLE IV
COMPARISON WITH PRIOR APPROACHES (TEST)

Method	Dice	IoU	Test Dataset
Midsem code (RF, 24 features)	0.67	0.55	Same
This version (20 features)	0.751	0.641	Same

V. CONCLUSION AND FUTURE WORK

This report outlined a complete classical machine learning pipeline to run ultrasound muscle segmentation, the outputs of which are attained through a series of sequential experiments, which culminated in this version. The final pipeline obtains a mean Dice of 0.751 and Remove of 0.641 on the withheld test set and the approximations of ACSA are correlated between estimates of ACSA and clinical ground truth with $r = 0.680$.

The biggest design choices were: application of a negative sampling concentrated at the boundary (2000 samples per image), enhancement of intensities before the extraction of the features, Gaussian smoothing of the probability maps before thresholding and adjustment of the value that defines decision

threshold to the validation set. The fact that the spatial feature had been ablated indicated that the spatial feature is not just exploiting positional shortcuts - appearance features are conceptually doing the correct job and spatial priors are not so significant.

Its main limitations include: the pipeline performs poorly at oblique scan angle and thick fat layers; the pixel-to-area calibration is estimated as it uses statistics across the dataset as opposed to imaging metadata per image, and the size of the training set is small, which hampers generalisation to out-of-sample imaging conditions.

In the future, a number of obvious improvements can be made:

- (i) instead of the pixel-independent Random Forest, a structured prediction model (such as a Conditional Random Field) can be used to enforce more smooth boundary changes;
- (ii) the hand-crafted features can be complemented with learned filter responses obtained using a small convolutional network;
- (iii) the pipeline can be tested on other muscles and acquisition protocols to understand how well it generalises

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APPENDIX: RESULT FIGURES

The following figures show representative segmentation outputs from this version on validation and test images.

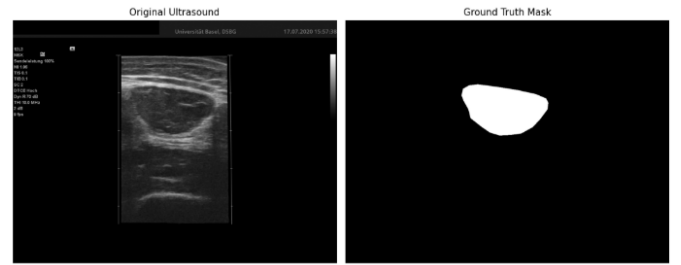


Fig. 2. Validation: rectus_img_23.tif

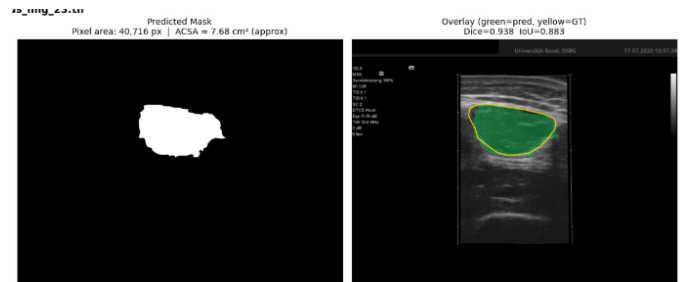


Fig. 3. rectus_img_23.tif

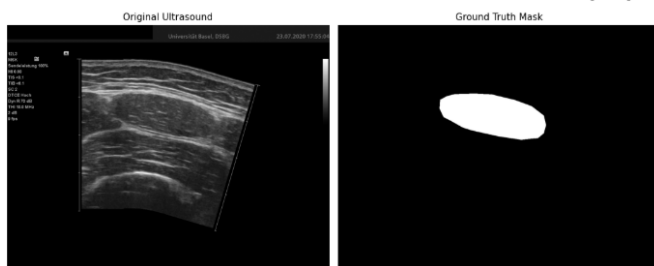


Fig. 4. Validation: Validation: rectus_img_346.tif

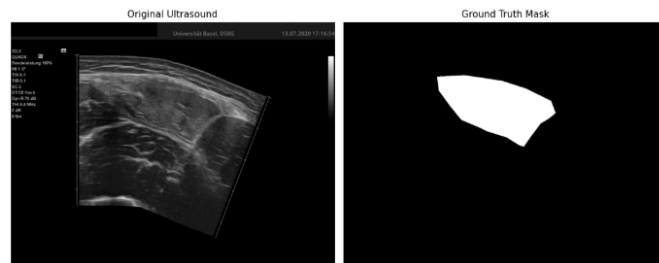


Fig. 8. Test: rectus_img_35.tif



Fig. 5. Validation: rectus_img_346.tif

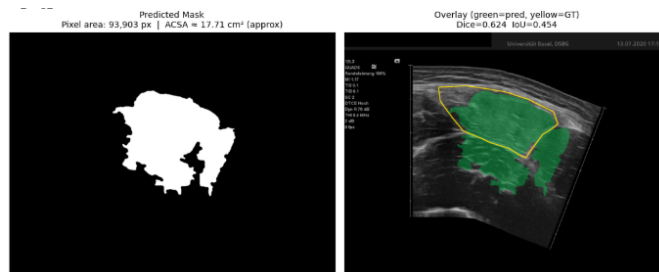


Fig. 9. Test: rectus_img_35.tif

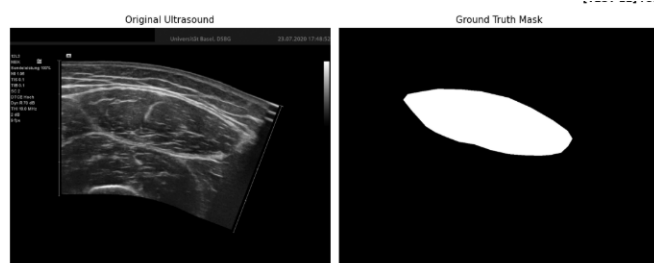


Fig. 6. Test: rectus_img_212.tif



Fig. 7. Test: rectus_img_212.tif

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